Connection Forms

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Abstract. This section transitions from curve theory to general frame fields by highlighting the core conceptual utility of the Frenet formulas, which derive their power by expressing the rates of change of the apparatus elements (T', N', B') strictly in terms of the basis vectors themselves to define curvature and torsion. This structural idea is then generalized to an arbitrary frame field E_1, E_2, E_3 on $\mathbf{E}^3$. By systematically expressing the covariant derivatives of these basis vector fields as linear combinations of the frame itself, a generalized framework is established for analyzing intrinsic geometric properties.

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1. Connection Forms

To express the covariant derivatives of an arbitrary frame field E_1, E_2, E_3 with respect to a tangent vector $\mathbf{v}$ at $\mathbf{p}$ in terms of the frame itself, we write them as a linear combination:

$$\begin{aligned}
\nabla_{\mathbf{v}} E_1 &= c_{11}E_1(\mathbf{p}) + c_{12}E_2(\mathbf{p}) + c_{13}E_3(\mathbf{p}) \\
\nabla_{\mathbf{v}} E_2 &= c_{21}E_1(\mathbf{p}) + c_{22}E_2(\mathbf{p}) + c_{23}E_3(\mathbf{p}) \\
\nabla_{\mathbf{v}} E_3 &= c_{31}E_1(\mathbf{p}) + c_{32}E_2(\mathbf{p}) + c_{33}E_3(\mathbf{p})
\end{aligned}$$

By exploiting the orthonormal properties of the frame field, we apply an orthonormal expansion to solve for these coefficients directly:

$$c_{ij} = \nabla_{\mathbf{v}} E_i \cdot E_j(\mathbf{p}) \quad \text{for } 1 \leq i, j \leq 3$$

Because these scalar coefficients c_{ij} depend entirely on the specific input tangent vector $\mathbf{v}$, we transition to a more functional notation:

$$\omega_{ij}(\mathbf{v}) = \nabla_{\mathbf{v}} E_i \cdot E_j(\mathbf{p}), \quad (1 \leq i, j \leq 3)$$

Here, each ω_{ij} represents a real-valued function (a 1-form) defined on all tangent vectors.

Lemma 1.1. *Let E_1, E_2, E_3 be a frame field on $\mathbf{E}^3$. For each tangent vector $\mathbf{v}$ to $\mathbf{E}^3$ at the point $\mathbf{p}$, let*

$$\omega_{ij}(\mathbf{v}) = \nabla_{\mathbf{v}} E_i \cdot E_j(\mathbf{p}), \quad (1 \leq i, j \leq 3)$$

Then each ω_{ij} is a 1-form, and it satisfies the skew-symmetry property:

$$\omega_{ij} = -\omega_{ji}$$

These 1-forms are called the ***connection forms*** *of the frame field E_1, E_2, E_3.*

Remark 1.2. The definition $\omega_{ij}(\mathbf{v}) = \nabla_{\mathbf{v}} E_i \cdot E_j(\mathbf{p})$ provides a direct geometric interpretation:

- **Rotational Interpretation:** The scalar value $\omega_{ij}(\mathbf{v})$ represents the *initial rate at which the basis vector E_i rotates toward the basis vector E_j* as the point $\mathbf{p}$ moves along the direction of the tangent vector $\mathbf{v}$.
- **Global Information:** Since each ω_{ij} is a differential 1-form, these matrix components package and track this rotational velocity structural data across all possible tangent vectors simultaneously throughout $\mathbf{E}^3$.

Theorem 1.3. *Let ω_{ij} $(1 \le i, j \le 3)$ be the connection forms of a frame field E_1, E_2, E_3 on $\mathbf{E}^3$. Then for any vector field V on $\mathbf{E}^3$:*

$$\nabla_V E_i = \sum_j \omega_{ij}(V) E_j, \quad (1 \le i \le 3)$$

*These identities are called the **connection equations** of the frame field E_1, E_2, E_3.*

When $i = j$, the skew-symmetry condition $\omega_{ij} = -\omega_{ji}$ gives $\omega_{ii} = -\omega_{ii}$, which implies:

$$\omega_{11} = \omega_{22} = \omega_{33} = 0$$

This condition reduces the nine 1-forms ω_{ij} down to essentially only three independent components: ω_{12}, ω_{13}, and ω_{23}. These components can be elegantly arranged as the entries of a skew-symmetric matrix of 1-forms:

$$\omega = \begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix} = \begin{pmatrix} 0 & \omega_{12} & \omega_{13} \\ -\omega_{12} & 0 & \omega_{23} \\ -\omega_{13} & -\omega_{23} & 0 \end{pmatrix}$$

By substituting these matrix properties back into the general connection equations (Theorem 7.2), we get the expanded form:

$$\begin{aligned} \nabla_V E_1 &= & \omega_{12}(V)E_2 + \omega_{13}(V)E_3 \\ \nabla_V E_2 &= -\omega_{12}(V)E_1 + & \omega_{23}(V)E_3 \\ \nabla_V E_3 &= -\omega_{13}(V)E_1 - \omega_{23}(V)E_2 & \end{aligned}$$

This system reveals an obvious structural relation to the classical **Frenet formulas** from curve theory:

$$\begin{aligned} T' &= & \kappa N & \\ N' &= -\kappa T + & & \tau B \\ B' &= & -\tau N & \end{aligned}$$

- **Vanishing Terms:** The missing $\omega_{13}(V)E_3$ and $-\omega_{13}(V)E_1$ terms in standard curve theory are a structural artifact of how the Frenet frame is constructed. Because N is chosen to align perfectly with the direction of T', the rate of change involves no component along the binormal vector B.
- **Dimensionality & Geometry:** While curvature κ and torsion τ only track geometric changes along a one-dimensional path (the curve's tangent T), an arbitrary frame field E_1, E_2, E_3 must account for directional variations throughout three-dimensional space $\mathbf{E}^3$. This necessitates the use of **1-forms** rather than static scalar functions.

The Attitude Matrix. To systematically compute the connection forms for any given frame field, we perform an orthonormal expansion of the frame vector fields E_i relative to the natural Euclidean frame fields U_1, U_2, U_3:

$$\begin{aligned} E_1 &= a_{11}U_1 + a_{12}U_2 + a_{13}U_3 \\ E_2 &= a_{21}U_1 + a_{22}U_2 + a_{23}U_3 \\ E_3 &= a_{31}U_1 + a_{32}U_2 + a_{33}U_3 \end{aligned}$$

The scalar entries are real-valued functions on $\mathbf{E}^3$ determined by the pointwise inner product:

$$a_{ij} = E_i \cdot U_j$$

Collecting these functions into a single matrix yields the **attitude matrix** A:

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Because the fields E_i and U_j are both strictly orthonormal, the attitude matrix A is an orthogonal matrix at every point $\mathbf{p}$, satisfying:

$$A^{-1} = A^T$$

Applying the exterior derivative to each component function of the matrix yields the differential matrix of 1-forms:

$$dA = (da_{ij})$$

Theorem 1.4. *If $A = (a_{ij})$ is the attitude matrix and $\omega = (\omega_{ij})$ is the matrix of connection forms of a frame field E_1, E_2, E_3, then*

$$\omega = dA\, A^T \quad \textit{(matrix multiplication)}$$

or, equivalently, in component notation:

$$\omega_{ij} = \sum_k a_{jk}\, da_{ik} \quad \textit{for } 1 \le i, j \le 3$$

REFERENCES

[1] B. O'Neill, *Elementary Differential Geometry*, Revised 2nd ed. Boston, MA: Academic Press, 2006.